

Wireless network technologies to be utilized for autonomous farming and crop cultivation in Albania

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Abstract- Using more and more internet enabled devices in our daily lives has brought significant growth to different industries around the world. Nowadays it has become very simple to remotely control multiple devices in order to work more efficiently. Given that agriculture seems to have a lot of potential for growth in Albania, the implementation of wireless technologies would certainly accelerate this growth. This paper reviews different technologies in order to find a viable way to improve farming and crop cultivation in Albania.

Keywords--- *monitoring system, Internet of Things (IoT), sensor, data, surveillance.*

I. INTRODUCTION

The Republic of Albania is a small country located on the Balkan Peninsula, along the Adriatic and Ionian sea. One of the largest and most important sectors in Albania is agriculture. The agriculture in Albania represents 18.9% of the GDP, which is significantly larger compared to other countries in the European Union where the agriculture sector represents around 2% of GDP (INSTAT 2019). [1]

The favorable climate, fertile land and access to a willing and competitive workforce, make Albania's farming look very promising for the future. However, generally there has not been much of an interest in investing in this sector, due to issues such as migration from rural areas, weak organization of the farmers, poor technological and infrastructure level, low level of development in the processing industry, etc. From an engineering point of view, we can think of a few ways to incorporate modern technology into traditional Albanian farming and cultivation, in order to help the sector grow and develop even further. This is why it's necessary to adapt the utilization of wireless technologies, as it would significantly improve Albanian agriculture on many aspects, such as organization and planning, monitoring of

the crops and farms, increasing production efficiency, etc.

Currently, this sector is dominated by small local producers, and every year there are about 2200 new enterprises operating in the agro-business, which causes fragmentation of the land. The monitoring of these lands will be done using devices connected by the Internet of Things, which will have a wide range connection, depending on the specific terrain. There will be GPS devices keeping track of the machinery, so that you can keep track of your assets and also ensure a good workflow. The wireless monitoring system will provide of one's assets and properties, and farmers will also have the opportunity to better observe how their crops are being affected by the environment (i.e.: humidity, soil PH level, etc.). To operate these devices a lot of energy will be needed, preferably an affordable and eco-friendly power source.

II. INTERNET ENABLED MONITORING SYSTEM

Implementing an autonomous monitoring system through wireless technologies might be the best way to improve Albanian agriculture. The system will use GPS tracing devices to keep track of tractors and machinery, there will be cameras to provide security and also sensors to observe weather conditions, in order to avoid crop loss. The data collected will help to better organize the agro-business. There are also plenty of online soft-wares that can connect through the wireless devices in order to collect data and make the work of farmers easier (such as TerraStar). For instance, these monitoring soft-wares can calculate fuel consumption, optimize the route and trace assets within a specified area. We can utilize this to create a highly-efficient and precise monitoring farming system. There will be a lot of data loading, so it is necessary to have several wireless sensors

available, which should be able to transfer data through a fast, encrypted network. This data will be handled through the Internet of Things (IoT), which will make the system accessible by mobile or computer. We need an IoT gateway device for the *edge computing*. The recommended device is “GL-X300B 4G LTE VPN Industrial Wireless Gateway, External Full-Band 4G Antennas, DNS Encryption, 2.4 GHz frequency”. [2] [3]



Figure 1 GL-X300B (Collie) 4G LTE VPN Industrial Wireless Gateway

It is an affordable device, with easy configuration, which will help implement the remote device management system. The data will be stored in the Internet Cloud where it can be easily accessed and analyzed.

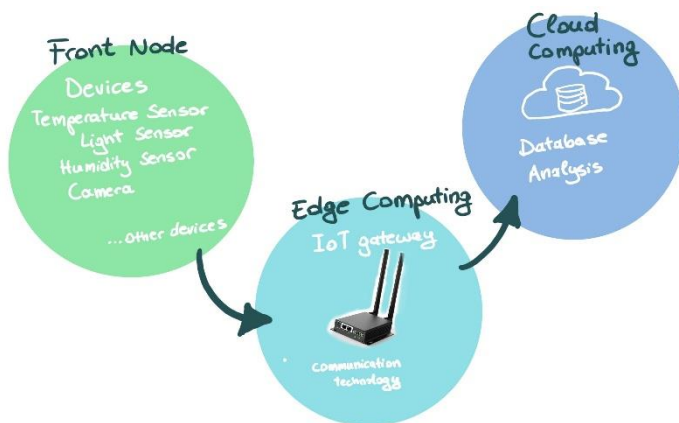


Figure 2 IoT system structure

III. TRACKING ASSETS

Industrial machinery is most certainly used across agro-businesses across Albania, so it is essential for business owners to keep good track of them.

The device selected for this task is “IoT Real-time GPS Tracking Device for Car NB-IoT Cat-M 4G Small Hidden GPS Locator for Vehicle”. This

device could easily be incorporated into the IoT monitoring system to make tracking easier.



Figure 3 IoT Real-time GPS Tracking Device

Tracking assets on real-time will ease the workflow of manually controlled machinery by optimizing the route of tractors and such. Different algorithms can be used to calculate the best trajectory for the machines to take in order to have maximized efficiency. Considering that most tractors plow around 3 hectares per day, and that the average Albanian farm is 0.5 hectares, we would need an average of 1 GPS tracking device per farm.

IV. MONITORING CONDITIONS FOR CROP GROWTH

As mentioned above, the wireless system will be using different sensors to monitor weather conditions (such as: temperature, humidity, soil PH, sunlight, irrigation, etc.) and other parameters. The IoT sensors will be measuring the parameters in real-time. If any abnormal conditions are detected, the alert will be sent to farmers immediately in order to avoid crops loss. Furthermore, farmers will be able to use the collected data to analyse the environment and deduce which crops are more suitable for certain seasons, in order to increase productivity.

The sensors will be programmed through arduino boards and set up across the whole land.

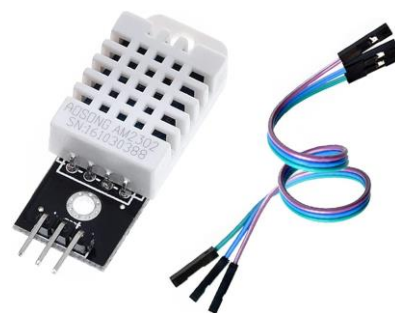


Figure 4 Digital Temperature And Humidity Sensor Module

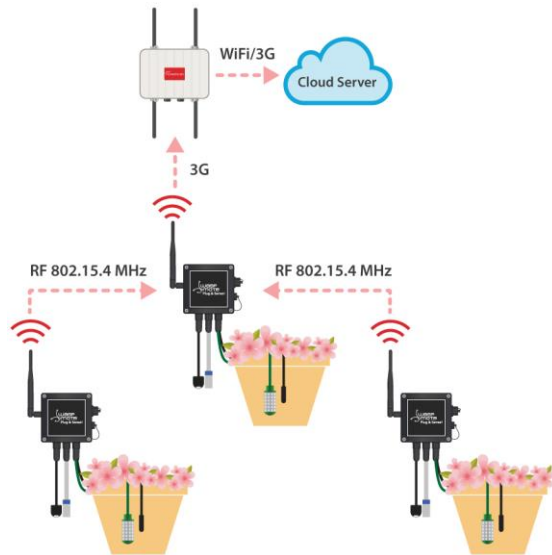


Figure 5 Collecting environmental data from sensors and analyzing it in the Cloud

V. CUSTOMIZING A WEBSITE FOR ALBANIAN FARMERS

As mentioned above, there are multiple soft-wares available for precision agriculture and organizing your agro-business. However, unfortunately, these soft-wares would be incompatible with the average local farmer in Albania. Most workers in this sector, probably don't know any English and will find it hard to use a complicated application. A more simplified, and customized website will be programmed to help farmers in Albania specifically. It will be accessible by all farmers in Albania and it will be in Albanian language. All of the data collected from the sensors and other devices will be accessible by the farmers through this website. There could also be different algorithms implemented into this website, for instance, you could find the most efficient trajectory for a tractor through mathematical calculations, which the website would do automatically.

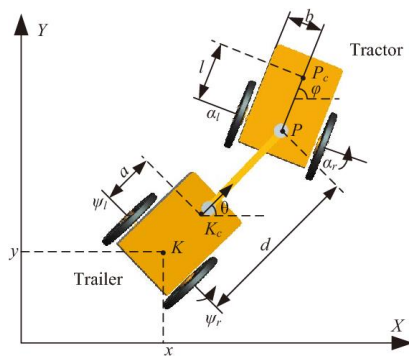


Figure 6 Calculations will be done for maximum efficiency

Furthermore, storing all of the data collected by sensors into the website will make it easier to analyze. Analyzing this information will result in saving time, improving decision making, reducing crop loss and increasing productivity.

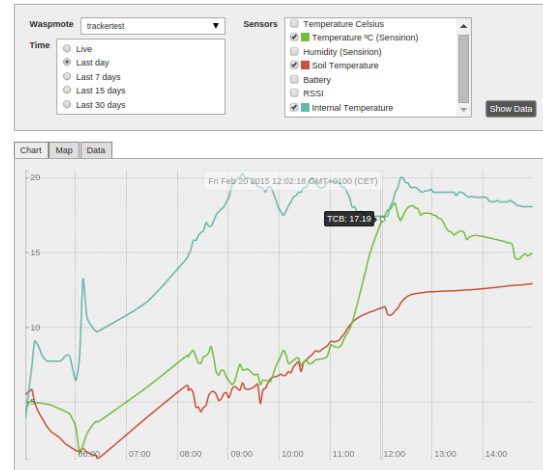


Figure 7 A reference of what analyzing data in the website might look like

VI. VISUALLY MONITORING LARGER AREAS

Apart from changing weather conditions, thieves and animal attacks might pose a threat to the farm as well. Which is why there will also be cameras and heat sensors linked to the monitoring system. Even in quieter rural areas, monitoring the land through cameras will save the farmers a lot of time. For monitoring livestock, regular cameras will be installed, whereas for monitoring the fields, it is more efficient to use drones. The utilization of drones is one of the most preferable solutions for monitoring crops, as it makes easier to cover larger areas. Given that a drone can cover more than 1.2 hectare in one flight, and more than 60 hectares in a day[9], and also that the average Albanian farm is around 0.5 hectares, this is most certainly an ideal solution for farmers who don't have a lot of income.



Figure 8 Carbon Fiber Mini Drones with Camera for Agriculture Aerial

The recommended cameras for surveillance, which can be referenced at the end of the paper, feature night vision, so they're ideal for monitoring and security purposes, and they're also solar powered to save energy. Suppose the average Albanian farm will need around 6 cameras for surveillance. Since we wouldn't be able to monitor the whole land with static cameras, they will be placed in key areas where surveillance is needed. For instance, if there's livestock in the farm, monitoring them with cameras is essential, and so on.

We will also be utilizing thermal cameras, because it is important to detect temperature changes in order to avoid any fires. Thermal cameras will be very handy because they can also detect movement of animals in the night, and also during hot summer days they will be used to monitor the workers, so that the farmers can be cautious about their employees getting heat exhaustion, or hypothermia in bleak winter days.

VII. POWER SOURCES

Even if farmers invest in modern wireless systems and sustainable devices to grow their business, one of the largest costs still comes from power consumption. Which is why it is necessary to find a reliable power source, which is affordable and efficient. Due to a favorable climate, Albania has around 300 sunny days in a year[1], so it only seems reasonable to use solar energy, which is both economic and eco-friendly.



Figure 9 Albania's cost-effective solar and wind potential is estimated at more than 7 gigawatts (GW)

VIII. COSTING INFORMATION

Table on the side shows the product costing as of 29th January, 2022.

Product Name	Quantity	Unit Price	Total Price
Solar Panel	12	486.78 \$	5,841.36 \$
Solar Classic 150 MPPT Charge Controller	1	644.00 \$	644.00 \$
G L-X300B (Collie) 4G LTE VPN Industrial Wireless Gateway	1	99.0 \$	99.0 \$
GPS Tracker for Vehicles[5]	1	49.99 \$	49.99 \$
Digital Temperature And Humidity Sensor Module[6]	30	14.49 \$	434.7 \$
Wi-Fi Water Level/Sump Monitor Sensor with Buzzer[12]	2	99.0 \$	198.0 \$
Songhe Capacitive Soil Moisture Sensor Module [11]	50	11.99 \$	599.5 \$
Carbon Fiber Mini Drones with Camera	1	2980.0 \$	2980.0 \$
Security Camera Wireless Outdoor, Solar Powered WiFi System, Pan Tilt, 2K Night Vision, 2-Way Talk	6	159.99 \$	959.94 \$
		Total Amount	11,806.49 \$

CONCLUSION

Implementing wireless technologies into Albanian agriculture in accordance with this paper, will ensure the further development of this sector in Albania. Modern technologies and IoT devices were made to improve our lives and develop different industries. The utilization of wireless technologies will certainly optimize the workflow of local Albanian farmers, ease their work and help them gather the necessary intel in order to increase productivity and therefore increase their income. Albania has already provided its locals with a variety of natural resources and good climate, so it is our chance to use our man made technologies in order to make the most out of what nature has already gifted us.

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